MD MUJAHIDUL ISLAM MRIDHA

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SUMMARY

Software engineer specializing in full-stack development, backend systems, and AI-powered applications. Experienced in building production web and mobile products, designing microservices, integrating payment systems, and developing retrieval-augmented generation (RAG) solutions. Focused on reliable software, maintainable architecture, and shipping end-to-end features.

WORK EXPERIENCE

Software Engineer | Antimattr Labs *July 2025 – September 2026*

- Restructured a multi-tenant gamified trivia SaaS platform into NestJS and MongoDB microservices, enabling teams to develop and release independently and in parallel, which shortened delivery time.
- Fixed leaderboard and payment concurrency issues by applying caching and atomic writes, improving platform reliability.
- Integrated bKash, BDApps, and EPS payment gateways; the bKash integration operates with very few transaction errors.
- Performed extensive debugging and a full UI overhaul of the platform in Astro.
- Built an Appsmith back-office tool for staff to manage tenants, questions, quizzes, and tournaments, track daily player insights, and generate reports.
- Developed a nearest-masjid finder, OneSignal push notifications, a multi-voice audiobook pipeline, and a FastAPI and LangChain RAG system that generates accurate math explanations.
- Worked in agile sprints, participated in code reviews, and supported GitOps deployments using Docker and Kubernetes.

Software Engineer | Fleeting Trails *June 2024 – June 2025*

- Independently designed and built a full-stack hall-booking platform using Laravel and React, covering bookings, payments, reporting, and accounting, replacing a paper-based process with a digital workflow.
- Developed features for a law-learning platform in Laravel and implemented load balancing and performance optimization to keep it stable under concurrent quiz and live-session traffic.

PROJECTS

Slashed | .NET, Flutter | Web and Android | slashed-app-web.vercel.app

- Built an offers-discovery app where users create category-based buckets and receive notifications when matching nearby items go on offer; supports browsing shops and listed items within a radius and searching products ranked by price.

Talk2 | Next.js, Go | Web | talk2-ui.vercel.app

- Developed a social platform centered on moment-to-moment interactions, with user security considered in the product design.

AiBit | Flutter, .NET | Web and Android | web-phi-ruddy-97.vercel.app

- Built a habit-tracking app that generates AI-powered personal reflections based on users' habit progress.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, C#, Go, Dart, PHP, C, SQL, MQL, RISC-V Assembly
- **Frontend and mobile:** Next.js, React, Astro, React Native, Expo, Flutter, Tailwind CSS, Zustand, TanStack Query
- **Backend and APIs:** NestJS, Node.js, FastAPI, .NET 8, Laravel, REST APIs, JWT, RBAC, HMAC, BullMQ, MongoDB, PostgreSQL, Redis, DragonflyDB, microservices architecture, algorithms
- **AI/ML:** PyTorch, TensorFlow, LangChain, Ollama, Hugging Face Transformers, OCR, RAG, ChromaDB
- **Cloud and DevOps:** Docker, Kubernetes, GitOps, FluxCD, Cloudflare Workers, S3/MinIO, GitHub Actions, Railway, Git
- **Integrations:** SSLCommerz, EPS, BDApps, Appwrite, OneSignal, FCM, OpenStreetMap, SMS gateways, Remotion
- **Engineering practices:** Agile methodologies, code review, unit testing, performance optimization, technical documentation

RESEARCH

Synthetic Image Detection Using CNNs. Detecting pixel-level and structural artifacts in AI-generated images.

Efficient Time-Conditioned Label Diffusion with Adaptive Local-Global Gating. Memory-efficient learning with label diffusion, time-conditioned denoising, and adaptive local-global aggregation.

ACTIVITIES

HR Executive, BRAC University Leadership Forum (2023 – 2026)

EDUCATION

BRAC University, B.Sc. in Computer Science, CGPA: 3.07 / 4.00 *2021 – 2026*